Examiner only

4. *(a)* Dilute hydrochloric acid reacts with sodium thiosulfate to make the products shown in the equation.

$$\boxed{\ \ ?\ \ }(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + SO_2(g) + S(s) + H_2O(l)$$

(i) Use the equation to work out the formula of sodium thiosulfate. [1]

Formula ....

(ii) The symbol (aq) in the equation tells us that the substances are aqueous. What is meant by this? [1]

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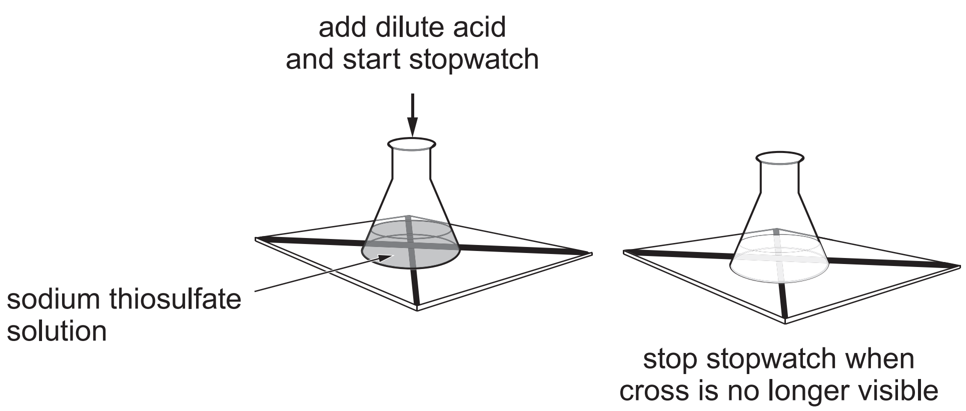
(iii) The rate of this reaction can be studied as shown in the diagram.



Use information **from the equation** to explain why the cross disappears. [2]

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(b) A student studied the effect of temperature on the rate of this reaction. He obtained the following results.

Temperature (°C)	Time taken for cross to disappear (s)			
	1	2	3	Mean
15	130	128	129	129
30	53	53	53	53
45	21	29	23	24.3
60	7	7	6	6.7

(i) Another student said that one of the mean values was incorrect. Identify the incorrect mean. Give your reasoning. [2]

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(ii) State what conclusion can be drawn about the effect of temperature on the rate of this reaction. Explain your conclusion using particle theory. [3]

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